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Functional Partial Least-Squares: Adaptive Estimation and Inference

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ABSTRACT
We study the linear regression model with a scalar response and a functional predictor, a canonical example of an ill-posed inverse problem. We show that the functional partial least-squares (PLS) estimator achieves convergence rates that are nearly minimax-optimal over a class of ellipsoids and propose an adaptive early-stopping procedure for selecting the number of PLS components. In addition, we develop a new test that detects parametric local alternatives. The test can be inverted to construct confidence sets for the functional slope parameter. Simulation results show that the estimator performs favorably relative to several existing methods, and that the proposed test has good power. We apply our methodology to evaluate the nonlinear effects of temperature on corn and soybean yields. We provide a Python software library, `fpls`, implementing our method. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

With the increasing availability of data, functional data analysis is now widely used across fields such as chemometrics, climate science, and economics. In this article, we study a linear functional regression model with a scalar response Y and functional predictor X:

$$Y = \int_0^1 \beta(s)X(s)\mathrm{d}s + \varepsilon, \qquad \mathbf{E}[\varepsilon X] = 0. \tag{1}$$

The primary objective is to estimate the functional slope β to predict the response Y. Both X and β lie in an infinite-dimensional Hilbert space, and the high dimensionality of β leads to an ill-posed inverse problem. Consistent estimation of the slope coefficient thus requires either a dimension-reduction technique or other form of regularization.

Two main approaches have been popular in the functional data analysis literature: penalization methods, for example, Cardot, Ferraty, and Sarda (2003), Li and Hsing (2007), Crambes, Kneip, and Sarda (2009), Yuan and Cai (2010), Cai and Yuan (2012), Florens and Van Bellegem (2015), and functional principal component analysis (PCA), for example, Cardot, Ferraty, and Sarda (1999), Ramsay and Silverman (2002), Cai and Hall (2006), Hall and Horowitz (2007). The PCA-based approach approximates β using a finite expansion over the leading principal components, which correspond to the eigenfunctions of the covariance operator. As noted by Jolliffe (1982), this method performs well when the response Y is primarily correlated with the leading principal components. Moreover, accurate recovery of the principal components requires sufficient separation between the eigenvalues of the covariance operator.

In this article, we explore an alternative method, functional partial least-squares (PLS). PLS is widely used in statistical learning, see Hastie, Tibshirani, and Friedman (2009), Frank and Friedman (1993); economics and finance, see Carrasco and Rossi (2016) and Kelly and Pruitt (2015); chemometrics, see Helland (1988) and Wold et al. (1984). However, it has been used less commonly in empirical functional data analysis. The method constructs components as linear transformations of the predictors X designed to maximize covariance with the response Y. As a result, fewer components are typically needed to achieve good predictive performance compared to PCA. Functional PLS was introduced by Preda and Saporta (2005) to address high-dimensional multicollinearity issues in the scalar-on-function linear model. Several interesting papers compare PCA and various functional PLS estimators using data-driven choices of the number of latent components in simulations; see Reiss and Ogden (2007), Krämer, Boulesteix, and Tutz (2008), Baíllo (2009), Aguilera et al. (2010), Febrero-Bande, Galeano, and González-Manteiga (2017), and Saricam et al. (2022). This literature concludes that the predictive ability of functional PCA and PLS approaches is similar, but functional PLS requires fewer components and provides more accurate estimates of the slope parameter than PCA.

However, the existing literature lacks supporting theory, which is challenging to obtain because the PLS estimator depends nonlinearly on the response variable and is computed iteratively. An important step toward the theoretical analysis of functional PLS was made by Delaigle and Hall (2012), who introduced an alternative PLS (APLS). Nevertheless, it remains unknown whether (a) functional PLS achieves nearly minimax-optimal convergence rates under weak conditions; (b) a rate-adaptive method to select the number of PLS components exists; and (c) inference for functional PLS can be conducted. Deriving an approximation to the distribution of the PLS estimator is particularly challenging because PLS is a nonlinear estimator

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converging at a nonparametric rate. Our paper fills these gaps in the literature and provides a comprehensive, rigorous theoretical analysis of functional PLS.

The article makes several original contributions. First, we derive convergence rates for estimation and prediction errors under a source condition. This condition measures the complexity of the problem through the so-called degree of ill-posedness and relates β to the spectral decomposition of the covariance operator. Our results do not require assuming a separation between adjacent eigenvalues, as is common for PCA (see, e.g., Hall and Horowitz 2007), and hold even in the presence of repeated eigenvalues. Second, we establish a lower bound on the minimax convergence rate and show that our estimator is nearly minimax-optimal. Because the optimal number of PLS components depends on the unknown degree of ill-posedness, we propose an adaptive early-stopping rule for selecting the number of components. We show that this single early-stopping rule yields a rate-optimal PLS estimator for *both* estimation and prediction errors with high probability. We also characterize how the selected number of components evolves with sample size under various scenarios.

Lastly, we develop a new test and accompanying confidence sets for PLS. We show that the test can detect alternatives converging at a parametric rate. Interestingly, while early stopping is crucial for estimation and prediction to prevent overfitting, it should be avoided for inference. An efficient iterative algorithm to compute the estimator is provided. Our simulation results reveal that the estimator outperforms several alternative methods combined with cross-validation in terms of estimation error and remains competitive in prediction. We also find that our test has excellent power with samples of as few as 100 observations.

To establish our theoretical results, we rely on the inverse problem literature and exploit the close connection between PLS and the conjugate-gradient method, as presented in Hanke (1995), Engl, Hanke, and Neubauer (1996), and Blanchard and Krämer (2016). To the best of our knowledge, this connection has not previously been established in the functional data analysis literature. Recently and independently of our work, Gupta, Sivananthan, and Sriperumbudur (2023) proposed an estimator in a reproducing kernel Hilbert space (RKHS) generated by a specified kernel and used the conjugate-gradient method to regularize the solution. Although their method is similar, the results are not directly comparable to ours. Their main result concerns the convergence rate of an estimator in an RKHS, established under different assumptions. In particular, they assume a polynomial decay rate of certain operator eigenvalues, while we do not require any decay assumptions; see also Gupta, Sivananthan, and Sriperumbudur (2025), who discuss optimal convergence rates with linear spectral regularization schemes under similar assumptions. Similarly, Lin and Cevher (2021) studied conjugate-gradient methods in the context of linear approximations to nonparametric regression with randomized sketches and Nyström sampling. While their results apply to general Hilbert spaces under stronger assumptions, they do not explicitly connect to the functional data analysis literature; for example, their simulations focus on linear approximations of $\mathbf{E}[Y|X] = f(X)$ with real-valued data $Y, X \in \mathbb{R}$ and Sobolev RKHS kernels. Neither paper establishes rate adaptivity or provides a practical early-stopping rule for selecting the number of PLS components.

To illustrate the practical relevance of our results, we apply our method to climate science. Using a fine-grained county-level dataset of U.S. crop yields and temperatures recorded over 70 years, we estimate the impact of temperature on crop yields. We find that the critical temperature at which annual crop yields begin to decay is around 30°C, consistent with Schlenker and Roberts (2009), who relied on highly parameterized least-squares estimators. Our method provides additional insights by examining how the temperature-effect curves have evolved over time. Interestingly, we find that the detrimental effects of high temperatures on corn and soybean yields have diminished over time, likely reflecting farmers' adaptive actions, including the use of more resilient crops and improved irrigation systems. However, this finding is not conclusive once statistical uncertainty is accounted for, given the small number of observed extreme temperatures.

The rest of the article is organized as follows. Section 2 introduces the functional regression model and the functional PLS estimator. Section 3 establishes the theoretical properties, including the convergence rates for estimation and prediction errors, the minimax lower bound, and the adaptivity of the early-stopping rule. Section 4 develops inference for PLS. Section 5 presents a Monte Carlo study. Section 6 discusses an empirical application to nonlinear temperature effects in agriculture. Section 7 concludes the article. The supplementary material provides all proofs, comparisons to alternative estimators, and additional simulation results. A replication package is also provided. Finally, we provide a Python library, `fpls`, available at *https://github.com/ababii/FPLS*, which implements the proposed methods.

2. Functional Regression and PLS Estimator

2.1. Functional Linear Regression

Throughout the article, we consider a generalized version of the functional linear regression model

$$Y = \langle \beta, X \rangle + \varepsilon, \qquad \mathbf{E}[\varepsilon X] = 0,$$

where $(Y, X) \in \mathbb{R} \times \mathbb{H}$, $\beta \in \mathbb{H}$ is the unknown functional slope coefficient, and $(\mathbb{H}, \langle ., . \rangle)$ is a separable Hilbert space with the induced norm $\|.\| = \sqrt{\langle ., . \rangle}$. The model in (1) corresponds to the Hilbert space of square-integrable functions, $\mathbb{H} = L_2[0, 1]$, with the norm induced by the inner product $\langle f, g \rangle = \int_0^1 f(s)g(s)\mathrm{d}s$.

If $\mathbf{E}[X] = 0$, the covariance restriction $\mathbf{E}[\varepsilon X] = 0$ implies that the slope coefficient $\beta \in \mathbb{H}$ solves the moment condition

$$r := \mathbf{E}[YX] = \mathbf{E}[(X \otimes X)\beta] =: K\beta, \tag{2}$$

where $r \in \mathbb{H}$ and $K : \mathbb{H} \to \mathbb{H}$ is a compact covariance operator with summable eigenvalues whenever $\mathbf{E}\|X\|^2 < \infty$. It is well known that the inverse operator K^{-1} is discontinuous and solving the equation $K\beta = r$ for β is an ill-posed inverse problem; see Carrasco, Florens, and Renault (2007), Engl, Hanke, and Neubauer (1996), and Hanke (1995).

Roughly speaking, there are two popular strategies to regularize such problems:

(a) replace K^{-1} with a continuous operator $R_\alpha(K)$ for some function $R_\alpha : \mathbb{R}_+ \to \mathbb{R}_+$ satisfying $\lim_{\alpha \to 0^+} R_\alpha(\lambda) = \lambda^{-1}$.

(b) solve the problem in a finite-dimensional subspace $\mathbb{H}_m \subset \mathbb{H}$, spanned by some fixed basis vectors $h_1, h_2, \ldots, h_m \in \mathbb{H}$.

Examples of (a) include Tikhonov regularization when $R_\alpha(\lambda) = (\alpha + \lambda)^{-1}$, the spectral cutoff when $R_\alpha(\lambda) = \lambda^{-1}\mathbf{1}_{\lambda \geq \alpha}$ and Landweber iterations. On the other hand, the estimators in group (b) often solve the empirical least-squares problem

$$\min_{b \in \mathbb{H}_m} \|\mathbf{y} - T_n b\|_n^2, \tag{3}$$

where $\|v\|_n^2 = v^\top v/n, v \in \mathbb{R}^n$ and we denote $\mathbf{y} = (Y_1, \ldots, Y_n)^\top$ and

$$T_n : \mathbb{H} \to \mathbb{R}^n$$
$$b \mapsto (\langle X_1, b\rangle, \ldots, \langle X_n, b\rangle)^\top$$

for an iid sample $(Y_i, X_i)_{i=1}^n$. The basis $(h_j)_{j=1}^m$ spanning $\mathbb{H}_m$ can be either fixed (e.g., Fourier, polynomials, splines, wavelets) or adaptively constructed from the data.

The data-driven bases are especially attractive since they can adapt to the features of the population represented by the data and can approximate the slope parameter $\beta \in \mathbb{H}$ more efficiently; see Delaigle and Hall (2012). The principal component analysis (PCA)[1] and the partial least-squares (PLS) are two widely used methods to construct adaptive bases in practice. The PCA basis is constructed by identifying the directions in $\mathbb{H}$ where X varies the most while the PLS basis is constructed in a supervised way taking into account the response variable as well. While the first m elements of the PCA basis $h_1, \ldots, h_m$ usually capture most of the variation of X, these are not necessarily the most important vectors for approximating β or predicting the response variable Y. It is easy to find empirical examples where some of the last few low-variance components *are* important; see Jolliffe (1982) who documented this issue using datasets from economics, climate science, chemical engineering, and meteorology.

2.2. PLS Estimator

The PLS estimator constructs a data-driven basis iteratively maximizing the covariance with the response variable Y; see Preda and Saporta (2005) who introduced it in the functional data analysis setting. The iterative nature of the estimator makes it difficult to analyze its statistical properties. This prompted Delaigle and Hall (2012) to formulate an alternative functional PLS that solves the problem in (3) over the so-called Krylov subspace

$$\mathbb{H}_m = \operatorname{span}\left\{\hat{r}, \hat{K}\hat{r}, \hat{K}^2\hat{r}, \ldots, \hat{K}^{m-1}\hat{r}\right\},$$

where

$$\hat{r} = \frac{1}{n}\sum_{i=1}^n Y_i X_i \qquad \text{and} \qquad \hat{K} = \frac{1}{n}\sum_{i=1}^n X_i \otimes X_i$$

are the estimators of r and K; see also Wold et al. (1984), Helland (1988), and Phatak and de Hoog (2002) for the link between PLS and Krylov subspaces.

In this article, we study a version of the PLS estimator with $m \geq 1$ components, denoted $\hat{\beta}_m$, characterized as a solution to the least-squares problem

$$\min_{b \in \mathbb{H}_m} \|T_n^*(\mathbf{y} - T_n b)\|^2$$

over the Krylov subspace $\mathbb{H}_m$. The least-squares objective function is weighted by the adjoint operator of T_n

$$T_n^* : \mathbb{R}^n \to \mathbb{H}$$
$$\phi = (\phi_1, \ldots, \phi_n)^\top \mapsto \frac{1}{n}\sum_{i=1}^n X_i \phi_i$$

and corresponds to minimizing the normal equations (the first-order conditions to the problem in (3)). Equivalently, $\hat{\beta}_m$ fits the empirical counterpart to the (2)

$$\min_{b \in \mathbb{H}_m} \left\|\hat{r} - \hat{K}b\right\|^2 \tag{4}$$

since $\hat{r} = T_n^*\mathbf{y}$ and $\hat{K} = T_n^* T_n$. Importantly, the PLS estimator formalized in (4) corresponds to the conjugate-gradient method applied to the self-adjoint operator $\hat{K}$, see Hestenes and Stiefel (1952), known for its excellent regularization properties; see also Hanke (1995) and Nemirovski (1986).[2] We provide a more detailed comparison between the two PLS estimators in the supplementary material, Section S.4. A related formulation of the PLS in reproducing kernel Hilbert spaces (RKHS) was recently studied in an independent work of Gupta, Sivananthan, and Sriperumbudur (2023) who focus on the estimation error only and impose assumptions different from ours. Our work can be seen as using a kernel naturally adapted to the data which is unknown in practice.

The estimator is uniquely defined for every $m \leq n_*$, where n_* is the number of distinct nonzero eigenvalues of $\hat{K}$; see Proposition S.1 in the supplementary material. It is also easy to see that for every $m \geq 1$, we have $\hat{\beta}_m = \hat{P}_m(\hat{K})\hat{r}$ for a polynomial $\hat{P}_m(\hat{K}) = \sum_{j=1}^m a_j \hat{K}^{j-1}$ with coefficients $\mathbf{a} := (a_1, \ldots, a_m)^\top$, where $\hat{P}_0 = 0$ and $\hat{\beta}_0 = 0$. The coefficient vector solves the system of m linear equations

$$\mathbf{K}\mathbf{a} = \mathbf{r},$$

where $\mathbf{K} := \langle \hat{K}^j\hat{r}, \hat{K}^k\hat{r}\rangle_{1 \leq j,k \leq m}$ and $\mathbf{r} := \langle \hat{K}^j\hat{r}, \hat{r}\rangle_{1 \leq j \leq m}$. From a practical point of view, it is more efficient to use an iterative conjugate-gradient algorithm that bypasses the (potentially unstable) matrix inversion with an iterative multiplication by the operator $\hat{K}$; see Algorithm 1 in Section 5.

3. Adaptive Estimation

In this section, we will show that the functional PLS estimator achieves the nearly optimal convergence rate on a class of ellipsoids. We consider an early-stopping rule to select the number of PLS components and show that it adapts to the complexity of the ellipsoid. Lastly, we study how rapidly the number of selected components increases with the sample size and make some comparisons to the PCA estimator.

[1] Using the PCA basis is also related to the spectral cutoff method described in (a).

[2] The conjugate-gradient method is one of the most efficient algorithms for solving high-dimensional systems of linear equations; see also (Nocedal and Wright 1999, chap. 5) and references therein.

3.1. Optimal Convergence Rates

Since the operators $K : \mathbb{H} \to \mathbb{H}$ and $\hat{K} : \mathbb{H} \to \mathbb{H}$ are self-adjoint and compact, by the spectral theorem

$$K = \sum_{j=1}^{\infty} \lambda_j v_j \otimes v_j \qquad \text{and} \qquad \hat{K} = \sum_{j=1}^{n} \hat{\lambda}_j \hat{v}_j \otimes \hat{v}_j,$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ and $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_n \geq 0$ are the eigenvalues of K and $\hat{K}$ and $(v_j)_{j=1}^{\infty}$ and $(\hat{v}_j)_{j=1}^{n}$ are the corresponding eigenvectors; see Kress (2014), Theorem 15.16. Note that the sample covariance operator $\hat{K}$ is a finite-rank operator with at most $n_* \leq n$ distinct nonzero eigenvalues.

For any bounded and measurable function $\phi : \mathbb{R}_+ \to \mathbb{R}_+$, we define functions of operators through their spectral decompositions:

$$\phi(K) := \sum_{j=1}^{\infty} \phi(\lambda_j) v_j \otimes v_j \qquad \text{and} \qquad \phi(\hat{K}) := \sum_{j=1}^{n_*} \phi(\hat{\lambda}_j) \hat{v}_j \otimes \hat{v}_j.$$

These definitions are commonly used in the inverse problems literature; see Engl, Hanke, and Neubauer (1996).

The following inequalities for the operator norm will often be used:

$$\|\phi(K)\|_{\text{op}} \leq \sup_{\lambda \in [0, \lambda_1]} |\phi(\lambda)| \qquad \text{and}$$
$$\|\phi(\hat{K})\|_{\text{op}} \leq \sup_{\lambda \in [0, \hat{\lambda}_1]} |\phi(\lambda)|, \tag{5}$$

where $\|A\|_{\text{op}} = \sup_{\|x\|=1} \|Ax\|$.

We next introduce several weak assumptions on the data-generating process.

Assumption 1. $(X_i, Y_i)_{i=1}^n$ are iid copies of (X, Y) with $\mathbf{E}[X] = 0$, $\mathbf{E}\|X\|^4 < \infty$, and $\mathbf{E}[\varepsilon^2 | X] \leq \sigma^2 < \infty$.

Assumption 1 imposes mild restrictions on the data-generating process. Note that $\mathbf{E}\|X\|^4 < \infty$ is satisfied when X is a Gaussian process in $\mathbb{H}$. It implies that K is a nuclear operator and, hence, compact.

Assumption 2. The operator $K : \mathbb{H} \to \mathbb{H}$ does not have zero eigenvalues.

Assumption 2 ensures that the slope parameter β is identified. If this assumption is violated, our results apply to the identified component of β within the orthogonal complement of the null space of K; see Babii and Florens (2025) and Engl, Hanke, and Neubauer (1996).

Assumption 3. For some $\mu, C > 0$, the slope parameter β and the operator K belong to the class

$$\mathcal{S}(\mu, C) = \Big\{ \beta \in \mathbb{H},\ K : \mathbb{H} \to \mathbb{H} : \quad \sum_{j=1}^{\infty} \frac{\langle \beta, v_j \rangle^2}{\lambda_j^{2\mu}} \leq C^2, \quad \sum_{j=1}^{\infty} \lambda_j \leq C \Big\}.$$

Assumption 3 describes the degree of ill-posedness in terms of the smoothness of β and the smoothing properties of the operator K. The parameter μ is known as the degree of ill-posedness. It restricts the rate of decay of the generalized Fourier coefficients $\langle \beta, v_j \rangle_{j \geq 1}$ relative to the eigenvalues of K. A larger value of μ means that it is easier to estimate the slope coefficient β; see also Carrasco, Florens, and Renault (2007). Recall also that the summability of eigenvalues holds whenever $\mathbf{E}\|X\|^2 < \infty$. Note that Assumption 3 is weaker than what is typically used to analyze the PCA estimators and does not require restricting the spacing between eigenvalues, see Hall and Horowitz (2007).

Consider now the so-called residual polynomial $\hat{Q}_m(\lambda) = 1 - \lambda \hat{P}_m(\lambda)$, deriving its name from the identity $\hat{r} - \hat{K}\hat{\beta}_m = \hat{Q}_m(\hat{K})\hat{r}$. It is known that the polynomial, $\hat{Q}_m$, has m distinct real roots, denoted $\hat{\theta}_1 > \hat{\theta}_2 > \cdots > \hat{\theta}_m > 0$. The sum of the inverses of these roots,

$$|\hat{Q}_m'(0)| = \sum_{j=1}^{m} \frac{1}{\hat{\theta}_j},$$

plays an important role in the analysis of the conjugate-gradient regularization; see Lemma S.4 in the supplementary material.

Our first result characterizes the convergence rate of the estimation and prediction errors of the PLS estimator.

Theorem 1. Suppose that Assumptions 1–3 are satisfied. Then for every $s \in [0, 1]$, we have

$$\left\| K^s(\hat{\beta}_m - \beta) \right\|^2 = O_P\Big(|\hat{Q}_m'(0)|^{2(1-s)} n^{-1} + |\hat{Q}_m'(0)|^{-2(\mu+s)} + |\hat{Q}_m'(0)|^{-2s} n^{-\mu \wedge 1} \Big),$$

provided that $|\hat{Q}_m'(0)| = O_P(n^{1/2})$.

Note that the last condition in Theorem 1 imposes that the number of components m does not increase too fast with the sample size and is not binding. In fact, it is optimal to have $|\hat{Q}_m'(0)| \sim n^{\frac{1}{2(\mu+1)}}$, in which case we obtain the following convergence rate

$$\left\| K^s(\hat{\beta}_m - \beta) \right\|^2 = O_P\left(n^{-\frac{\mu+s}{\mu+1}} \right).$$

When $s = 0$, this shows that the convergence rate of PLS in the Hilbert space norm is of order $n^{-\frac{\mu}{\mu+1}}$. On the other hand, when $s = 1/2$, we obtain the convergence rate of the out-of-sample prediction error, since

$$\mathbf{E}_X \langle X, \hat{\beta}_m - \beta \rangle^2 = \left\| K^{1/2}(\hat{\beta}_m - \beta) \right\|^2,$$

where $\mathbf{E}_X$ is taken with respect to X, independent of $(Y_i, X_i)_{i=1}^n$.

Remark 1. Note that the consistency of functional PLS has been established previously in Delaigle and Hall (2012) assuming that the eigenvalues are summable only which can be stated as $\mu = 0$. Characterizing the speed of convergence, however, requires more regularity with $\mu > 0$.

The following result shows that no estimator can achieve faster than $n^{-\frac{\mu+s}{\mu+1}} \log^{-b} n$ rate on the class $\mathcal{S}(\mu, C)$.

Theorem 2. For every $s \in [0, 1/2]$, there exists $A < \infty$ such that

$$\liminf_{n\to\infty} \inf_{\hat{\beta}} \sup_{(\beta,K)\in\mathcal{S}(\mu,C)} \Pr\left(\left\|K^s(\hat{\beta}-\beta)\right\|^2 \geq An^{-\frac{\mu+s}{\mu+1}} \log^{-b} n\right) > 0,$$

where $b > 2(\mu + s)$ and the infimum is over all estimators.

Therefore, we conclude that the PLS estimator $\hat{\beta}_m$ achieves the nearly optimal convergence rate on $\mathcal{S}(\mu, C)$, simultaneously for the estimation ($s = 0$) and prediction ($s = 1/2$) errors.

Remark 2. Note that if β is in $\mathcal{S}(\mu, C)$ with $\mu = 1/2$, then it also belongs to the RKHS generated by the covariance operator K. In this case, we show that the minimax-optimal rate of prediction ($s = 1/2$) is $O_P(n^{-2/3})$. The result does not seem to be comparable to Cai and Yuan (2012), since their minimax-optimal $O_P(n^{-2r/(2r+1)})$ rate is obtained under the additional assumption that the eigenvalues decay polynomially fast, that is $\lambda_j \sim j^{-r}$.[3] The value $r = 1$ is not feasible since the existence of the covariance operator holds under $\mathbf{E}\|X\|^2 < \infty$ which is equivalent to $\sum_{j=1}^{\infty} \lambda_j < \infty$. In contrast, our class $\mathcal{S}(\mu, C)$ with $\mu = 1/2$ does not require specifying the decay rate of eigenvalues.

3.2. Adaptive PLS Estimator

Next, we look at the adaptive PLS estimator, where the number of components is selected using the early-stopping rule described in the following assumption.

Assumption 4. We select $\hat{m}$ as the smallest $m \geq 0$ such that

$$\left\|\hat{r} - \hat{K}\hat{\beta}_m\right\| \leq \tau\sigma\sqrt{\frac{2\mathbf{E}\|X\|^2}{\delta n}}$$

for $\tau > 1$ and some $\delta \in (0, 1)$.

Assumption 4 states that the selected number of PLS components $\hat{m}$ equals the first nonnegative integer m such that the norm of the fitted "moment" is smaller than a certain threshold; see supplementary material, Section S.5 for a practical implementation of this early-stopping rule. Note that the number of selected components is finite since $\hat{m} \leq n_*$, where n_* is the number of distinct nonzero eigenvalues of $\hat{K}$; see Proposition S.1 in the supplementary material. In fact, the norm of the "residual" is zero for $m \geq n_*$ which corresponds to perfect fit.

The following result shows that the early-stopping rule in Assumption 4 is adaptive to the unknown degree of ill-posedness $\mu > 0$.

Theorem 3. Suppose that Assumptions 1–4 hold with $\delta \geq 1/n$. Then

$$\left\|K^s(\hat{\beta}_{\hat{m}} - \beta)\right\|^2 = O\left((\delta n)^{-\frac{\mu+s}{\mu+1}}\right)$$

with probability at least $1 - \delta$ for every $s \in [0, 1]$.

[3]Similar assumptions are also made in Gupta, Sivananthan, and Sriperumbudur (2023), Gupta, Sivananthan, and Sriperumbudur (2025), and Lin and Cevher (2021).

Taking $\delta_n = 1/\log n$ in Assumption 4, we obtain from Theorem 3 the convergence rate of the estimation and prediction errors of PLS with the number of components selected iteratively with the early-stopping rule:

$$\left\|K^s(\hat{\beta}_{\hat{m}} - \beta)\right\|^2 = O_P\left(\left(\frac{\log n}{n}\right)^{\frac{\mu+s}{\mu+1}}\right).$$

Therefore, the adaptive PLS achieves the nearly optimal convergence rate simultaneously for the estimation and prediction errors without knowing the degree of ill-posedness $\mu > 0$.

3.3. Number of Selected Components

In this section, we look at how rapidly the number of selected components in Assumption 4 increases with the sample size. First, we consider a somewhat conservative bound that does not impose any assumptions on the spectrum of the operator K.

Theorem 4. Suppose that Assumptions 1, 2, 3, and 4 are satisfied with $\delta \geq 1/n$ and $\mu \geq 1$. Then with probability at least $1 - \delta$

$$\hat{m} = O\left((n\delta)^{\frac{1}{4(\mu+1)}}\right).$$

Taking $\delta = 1/\log n$, we obtain from Theorem 4 that $\hat{m} = O_P\left((n/\log n)^{\frac{1}{4(\mu+1)}}\right)$. Next, we consider sharper estimates under additional assumptions imposed on the spectrum of the operator K.

Theorem 5. Suppose that Assumptions 1–4 are satisfied with $\delta \geq e/n$ and $\mu \geq 1$. Then with probability at least $1 - \delta$

(i) If $\lambda_j = O(j^{-2\kappa})$ for some $\kappa > 0$, then

$$\hat{m} = O\left((n\delta)^{\frac{1}{4(\kappa+1)(\mu+1)}}\right).$$

(ii) If $\lambda_j = O(q^j)$ for some $q \in (0, 1)$, then

$$\hat{m} = O\left(\log(n\delta)\right).$$

Theorem 5 shows that if the eigenvalues decay polynomially fast, then the selected number of components is $\hat{m} = O_P\left(n/\log n\right)^{\frac{1}{4(\kappa+1)(\mu+1)}}$ while in the case of the geometric decay, the number of selected components increases only logarithmically with the sample size. Therefore, the adaptive early-stopping rule will choose a smaller number of components if the eigenvalues of the operator K decay faster and vice versa.

4. Inference

4.1. A Hypothesis Test

We aim to test the null hypothesis against the alternative hypothesis:

$$H_0: \ \beta = b \qquad \text{versus} \qquad H_1: \ \beta \neq b,$$

for some known $b \in \mathbb{H}$. We are also interested in testing against a sequence of local alternative hypotheses

$$H_{1,n}: \ \beta = b + n^{-1/2}\Delta$$

for some $\Delta \in \mathbb{H}$.

Recall that our main result in Theorem 1 shows that

$$\left\|K^s(\hat{\beta}_m - \beta)\right\|^2 = O_P\left(n^{-\frac{\mu+s}{\mu+1}}\right), \qquad \forall s \in [0,1],$$

provided that the number of components m is properly selected. The convergence rate is $O_P(n^{-1})$ for $s = 1$ and it is slower than $O_P(n^{-1})$ for $s < 1$. This suggests that the following statistic

$$W_n = n\left\|\hat{K}(\hat{\beta}_m - b)\right\|^2$$

may have a well-defined asymptotic distribution under H_0.[4]

Let $V = \mathbf{E}[\varepsilon^2 X \otimes X]$ be the variance operator with spectral decomposition

$$V = \sum_{j=1}^{\infty} \omega_j \varphi_j \otimes \varphi_j$$

and let $Z_j \overset{\text{iid}}{\sim} N(0,1)$. The following theorem establishes the limiting distribution of W_n under the null and alternative hypotheses.

Theorem 6. Suppose that Assumptions 1 and 2 are satisfied and the PLS estimator is computed using m components such that $\|\hat{r} - \hat{K}\hat{\beta}_m\| = o_P(n^{-1/2})$. Then, under H_0,

$$W_n \xrightarrow{d} \sum_{j=1}^{\infty} \omega_j Z_j^2,$$

while under H_1, $W_n \xrightarrow{\text{a.s.}} \infty$. Moreover, under $H_{1,n}$,

$$W_n \xrightarrow{d} \sum_{j=1}^{\infty} \omega_j Z_j^2 + 2\sum_{j=1}^{\infty} \omega_j^{1/2} Z_j \langle \varphi_j, K\Delta\rangle + \|K\Delta\|^2.$$

Remark 3. The requirement $\|\hat{r} - \hat{K}\hat{\beta}_m\| = o_P(n^{-1/2})$ in Theorem 6 can be easily satisfied when the number of components m is sufficiently large. Indeed, we know that $\|\hat{r} - \hat{K}\hat{\beta}_m\| = 0$ for $m = n_*$, where $n_* \le n$ is the number of nonzero eigenvalues of the sample covariance operator $\hat{K}$, see supplementary material, Proposition S.1.

Remark 4. Theorem 6 illustrates an interesting phenomenon related to the optimal choice of the number of PLS components which amounts to early stopping of the conjugate-gradient method. While it is optimal to select a small number of components to prevent overfitting for estimation, this should be avoided for inference. In fact, it is evident from the proof of Theorem 6 that the overfitted PLS with zero fitting error leads to more accurate asymptotic approximation; see Bartlett et al. (2020) for benign overfitting in the linear regression.

Remark 5. The test has more power against the alternatives with larger values of

$$\|K\Delta\|^2 = \sum_{j=1}^{\infty} \lambda_j^2 \langle \Delta, v_j\rangle^2,$$

where $(\lambda_j, v_j)_{j=1}^{\infty}$ are the eigenvalues and eigenvectors of the operator K. Such alternatives are aligned with directions defined by the leading eigenvectors of K, that is directions in which it is easier to identify the slope function β.

Remark 6. Under homoscedasticity, $\mathbf{E}[\varepsilon^2|X] = \sigma^2$, we have $V = \sigma^2 K$. In this case, V has eigenvectors $\varphi_j = v_j$ with corresponding eigenvalues $\omega_j = \sigma^2\lambda_j$.

Theorem 6 implies that the test based on W_n has correct size under the null hypothesis and is consistent under the fixed alternative hypotheses which we state as a trivial corollary below. To that end, let $z_{1-\alpha}$ be the $1-\alpha$ quantile of

$$W := \sum_{j=1}^{\infty} \omega_j Z_j^2.$$

The test rejects H_0 if $W_n > z_{1-\alpha}$. Then, the following result holds:

Corollary 1. Suppose that the conditions of Theorem 6 are satisfied. Then under H_0, we have

$$\lim_{n\to\infty} \Pr(W_n > z_{1-\alpha}) = \alpha$$

while under H_1, we have

$$\lim_{n\to\infty} \Pr(W_n > z_{1-\alpha}) = 1.$$

Moreover, under $H_{1,n}$

$$\lim_{n\to\infty} \Pr(W_n > z_{1-\alpha}) \uparrow 1 \qquad \text{as} \qquad \|K\Delta\| \uparrow \infty.$$

It is also easy to show that the critical value $z_{1-\alpha}$ can be consistently estimated with simple nonparametric bootstrap based on the iid draws from the sample $(Y_i, X_i)_{i=1}^n$. Alternatively, one can simulate the asymptotic critical values using the eigenvalues $\hat{\omega}_j$ of the sample variance operator $\hat{V}$.

4.2. A Confidence Set

Inverting the test, we obtain a confidence set

$$C_{1-\alpha} = \{b \in \mathbb{H} : \ W_n(b) \le z_{1-\alpha}\},$$

where $z_{1-\alpha}$ is the quantile of order $1-\alpha$ of $W = \sum_{j=1}^{\infty} \omega_j Z_j^2$. Indeed, since $W_n \xrightarrow{d} W$, the confidence set has the coverage probability $1-\alpha$:

$$\begin{aligned}\lim_{n\to\infty} \Pr(\beta \in C_{1-\alpha}) &= \lim_{n\to\infty} \Pr(W_n(\beta) \le z_{1-\alpha}) \\ &= \Pr(T \le z_{1-\alpha}) = 1-\alpha.\end{aligned}$$

Since β is an infinite-dimensional object, computing this confidence set numerically requires finite-dimensional approximations which can be done as follows. Let $(h_j)_{j=1}^{\infty}$ be a basis of $\mathbb{H}$. Then we approximate $C_{1-\alpha}$ numerically by computing

$$C_{1-\alpha}^J = \left\{(b_1, \ldots, b_J) \in \mathbb{R}^J : \ W_n\left(\sum_{j=1}^{J} b_j h_j\right) \le z_{1-\alpha}\right\}.$$

For instance, in functional data analysis, we often have $\mathbb{H} = L_2[0,1]$ in which case, we can use the Fourier basis or Legendre polynomials. If $\mathbb{H} = L_2(\mathbb{R})$, the Hermite polynomials are a natural choice; see supplementary material, Section S.5 for numerical implementation and simulation results.

[4]The statistic is similar to Wald statistic which under homoscedasticity roughly corresponds to $s = -1/2$, hence, not converging at the parametric rate.

Algorithm 1: PLS algorithm for solving $\hat{K}\hat{\beta} = \hat{r}$.

Result: $\hat{\beta}_m$
Initialisation: $\hat{\beta}_0 = 0$, $d_0 = e_0 = \hat{r} - \hat{K}\hat{\beta}_0$;
for $j = 0, 1, \ldots, m-1$ **do**
1. Compute step size: $\alpha_j = \frac{\langle e_j, \hat{K}e_j \rangle}{\|\hat{K}d_j\|^2}$;
2. Update slope coefficient: $\hat{\beta}_{j+1} = \hat{\beta}_j + \alpha_j d_j$;
3. Update fitted moment: $e_{j+1} = e_j - \alpha_j \hat{K} d_j$;
4. Compute step size for the conjugate direction update: $\gamma_{j+1} = \frac{\langle e_{j+1}, \hat{K}e_{j+1} \rangle}{\langle e_j, \hat{K}e_j \rangle}$;
5. Update conjugate direction vector: $d_{j+1} = e_{j+1} + \gamma_{j+1} d_j$;

end

5. Monte Carlo Experiments

In this section, we conduct several Monte Carlo experiments to evaluate the finite-sample performance of the PLS estimator. We simulate iid samples $(Y_i, X_i)_{i=1}^n$ from the functional linear model

$$Y_i = \int_0^1 X_i(s)\beta(s)\mathrm{d}s + \varepsilon_i, \qquad \varepsilon_i \sim_{\text{iid}} N(0,1),$$

where the predictors X_i belong to the Hilbert space of square-integrable functions with respect to the Lebesgue measure, denoted $\mathbb{H} = L_2[0,1]$. The functional predictor is generated as

$$X_i(s) = \sum_{j=1}^{100} \sqrt{\lambda_j} u_j v_j(s), \qquad u_j \sim_{\text{iid}} N(0,1).$$

We specify the slope parameter $\beta \in L_2[0,1]$ and the spectrum $(\lambda_j, v_j)_{j\geq 1}$ correspond to one of the following three models:

- **Model 1:** $\beta(s) = \sum_{j=1}^{100} \beta_j v_j(s)$ with $\beta_j = 4j^{-2.7}$ and $\lambda_j = 2j^{-1.1}$ for all $j \geq 1$, $v_j(s) = \sqrt{2}\cos(j\pi s)$, $j \geq 2$, and $v_1(s) = 1$.
- **Model 2:** same as Model 1, but with $\beta_j = 4$ for all $j = 1, \ldots, 5$.
- **Model 3:** same as Model 1, but with $\lambda_j = 2$ for all $j = 1, \ldots, 5$.

All three models satisfy Assumption 3. Model 2 emphasizes the importance of the first five coefficients β_j, making the estimation and prediction tasks more challenging. Model 3 introduces repeated eigenvalues, causing the first five eigenvectors to be non-identifiable, although the slope parameter remains identifiable.

We compute the PLS estimator using Algorithm 1, which is numerically equivalent to the estimator given by (4); see Hanke (1995, Algorithm 2.1 and Proposition 2.1). This approach avoids direct inversion of the empirical operator $\hat{K}$ by using iterative multiplication, making it suitable for solving high-dimensional linear systems of the form $\hat{K}\hat{\beta} = \hat{r}$ with a symmetric matrix $\hat{K}$.

The integrals in inner products and the operator K are discretized using a simple Riemann sum approximation over a grid of $T = 200$ equidistant points in $[0,1]$.[5] The experiments feature 5000 replications, with each replication generating samples of size $n = 100$. For each simulation experiment and an estimator $\hat{\beta}$, we compute:

- The integrated squared error (ISE):
$$\mathrm{ISE}(\hat{\beta}) = \int_0^1 |\hat{\beta}(s) - \beta(s)|^2 \mathrm{d}s;$$
- the mean-squared prediction error (MSPE):
$$\mathrm{MSPE}(\hat{\beta}) = \frac{1}{n}\sum_{i=1}^n (Y_i - \langle X_i, \hat{\beta}\rangle)^2,$$
where the estimator $\hat{\beta}$ is computed from an auxiliary independent sample of size n.

We compare the performance of our functional PLS estimator with the number of components chosen using our early-stopping rule, against several alternative methods:[6]

1. A spline estimator, jointly selecting the spline degree and smoothing parameter via generalized cross-validation (GCV), following Crambes, Kneip, and Sarda (2009).
2. A principal component regression (PCA) estimator, where component selection is also done by GCV.
3. A reproducing kernel Hilbert space (RKHS) estimator using the same kernel as in Yuan and Cai (2010, eq. (16)).
4. The alternative PLS method from Delaigle and Hall (2012), selecting components through 5-fold cross-validation.

Note that Crambes, Kneip, and Sarda (2009), Proposition 2, establishes that the optimal rate of prediction for the smoothing spline estimator can be attained adaptively using generalized cross-validation. On the other hand, the kernel function employed in Yuan and Cai (2010) is not adaptively selected from the data. We are not aware of any adaptivity results for other competing estimators, including PCA and the alternative PLS estimator with cross-validation. In contrast, our method for selecting the number of PLS components is adaptive for both estimation and prediction according to Theorem 3.

Figure 1 summarizes the distributions of estimation errors (ISE) and prediction errors (MSPE) from 5000 simulations. The early-stopped PLS estimator achieves the lowest median estimation error, with consistently lower variability across all models. The penalized spline estimator performs effectively when coefficients decay rapidly but exhibits notably larger errors in Model 2, which involves high-frequency components. Nevertheless, it shows excellent predictive performance consistent with Crambes, Kneip, and Sarda (2009). The PLS estimator also demonstrates competitive predictive accuracy, although it does not yield the lowest median prediction error across all simulation designs. The RKHS estimator achieves good prediction performance but inferior estimation across all models. The alternative PLS estimator generally performs comparably to our proposed PLS approach but exhibits notably larger prediction errors for Models 2 and 3. These higher errors may be attributed to numerical instability associated with finite-precision arithmetic, as discussed in Delaigle and Hall (2012). Finally, PCA

[5] This gives a satisfactory approximation under weak assumptions. Alternatively, one could employ quadrature rules with fewer points to reduce computational load.

[6] See supplementary material, Section S.5, for more details on the practical implementation.

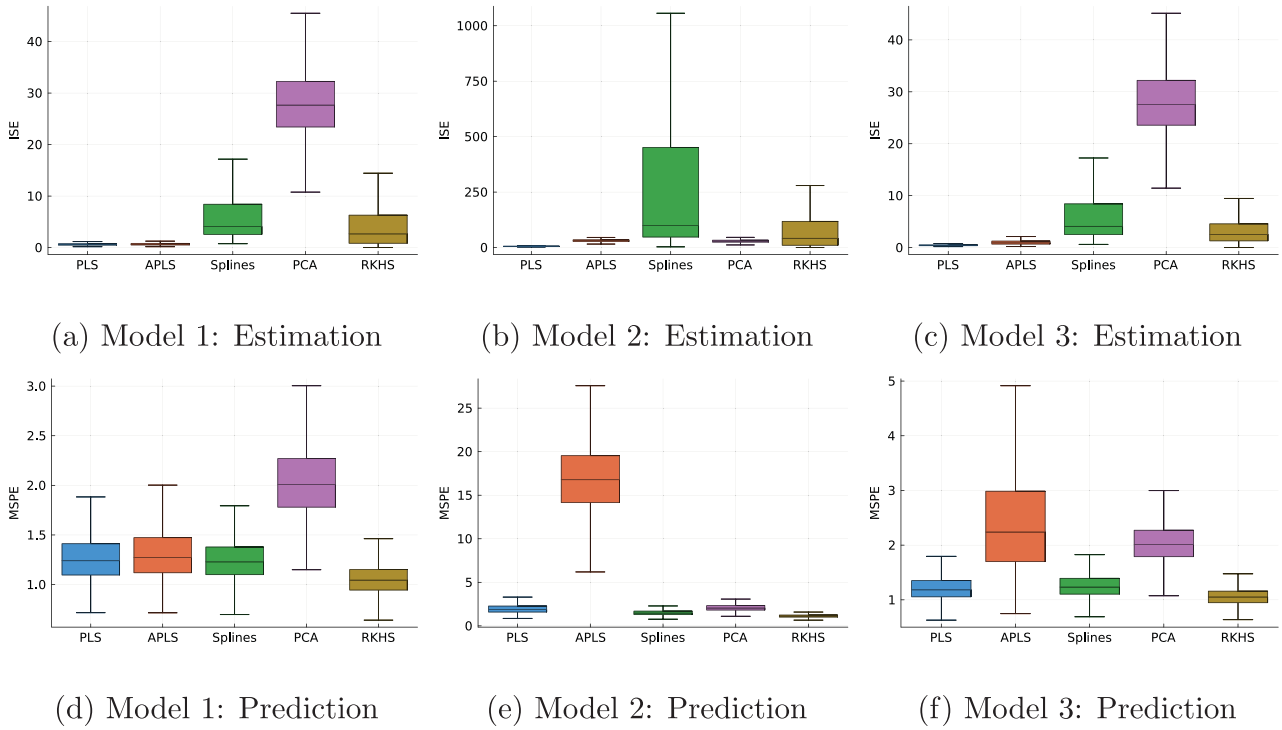


Figure 1. Estimation and prediction accuracy. Top row: integrated squared error (ISE) for the slope coefficient. Bottom row: mean squared prediction error (MSPE). Each boxplot summarizes 5000 Monte Carlo replications.

demonstrates relatively modest performance across all the considered models. Moreover, we find in simulations that the FPLS estimator outperforms FPCA using a smaller number of components offering a more parsimonious representation of the data; see also Frank and Friedman (1993) for related discussion and our theoretical results in supplementary material, Section S.3.

Next, we evaluate the performance of the proposed test. We first assess whether the asymptotic distribution provides a good approximation to the finite-sample distribution under the null hypothesis H_0. To ensure that the late stopping conditions of Theorem 6 are satisfied, we employ PLS with $m = 70$ components.[7] The results are presented in Figure 2.

For each of the three models, the top row in Figure 2 displays the finite-sample distribution of W_n (in blue) under H_0, overlaid with the simulated $\sum_{j=1}^{100} \omega_j Z_j^2$ (in orange) obtained from Theorem 6. The bottom row presents the corresponding Q–Q plots, comparing the empirical quantiles of W_n to the asymptotic quantiles of T. Overall, the simulated asymptotic distribution closely matches the finite-sample distribution, with only minor discrepancies observed in the extreme right tail.

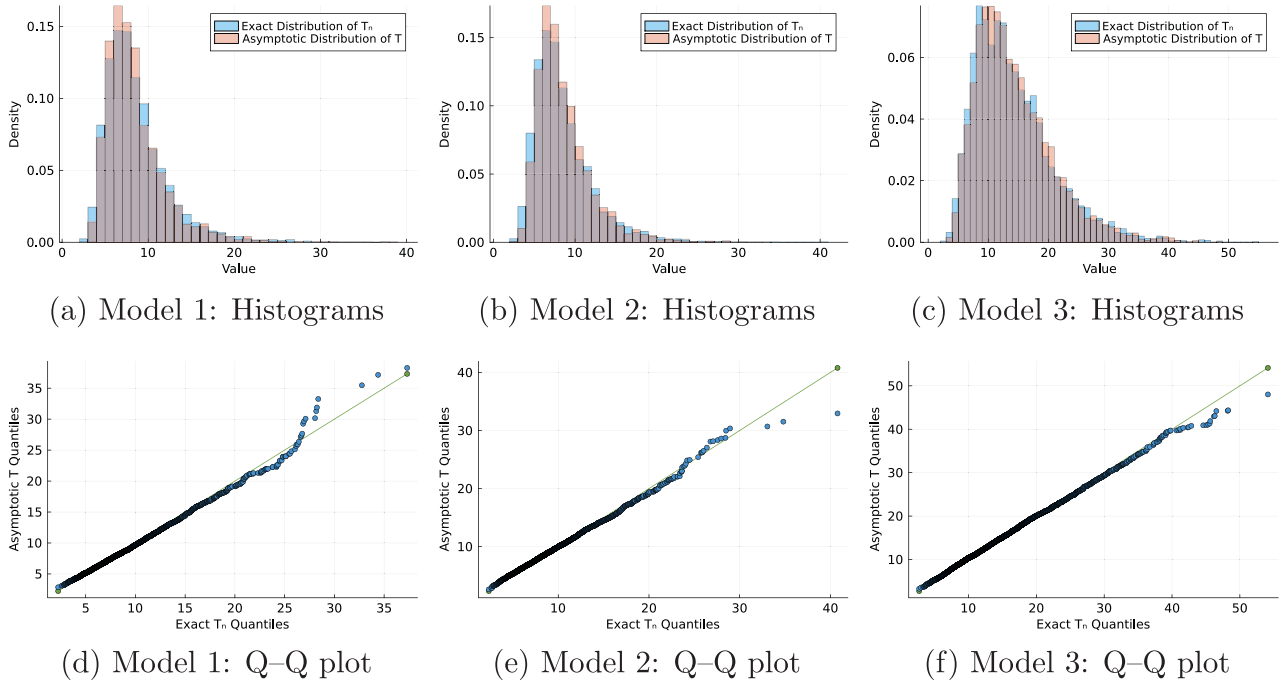
To study the finite-sample power of the test, we simulate the power curves corresponding to the local alternatives $\beta(s) + \delta s$ with deviations measured by the scale factor $\delta \in [-1, 1]$. $\delta = 0$ corresponds to H_0 while $|\delta| > 0$ to H_1. We also consider doubling the sample size from $n = 100$ to $n = 200$. The results are displayed in Figure 3, confirming that the test has more power once the null and the alternative hypotheses become sufficiently separated. The power also increases with the sample size as expected. Lastly, we provide additional simulation results in the supplementary material.

6. Nonlinear Temperature Effects in U.S. Agriculture

The global surface temperature has increased by 1.1°C above pre-industrial levels and could increase up to 3.6°C–4.5°C by the end of the century if current CO_2 emissions rise steadily according to the latest studies; see on Climate Change (IPCC). Global warming will likely lead to more frequent and severe heat waves, altered precipitation patterns, and intensified droughts. Of all major sectors, agriculture is arguably the most sensitive to climate change. While constituting a modest share of developed economies, it is vital for food security. Indeed, the intensified droughts could cause food shortages which in turn may exacerbate mass migration and violent conflicts. Some have argued that many regions are already warmer than optimal for agriculture in many parts of Asia, Africa, and Latin America; see Nordhaus (2013).

Determining the precise functional form of the relationship between crop yields and temperature has recently attracted lots of attention; see Schlenker and Roberts (2006, 2009) and Cui et al. (2024). We argue that the methodology used to estimate such nonlinear temperature effects can be understood as a functional linear regression, where the outcome Y_{it} is the log-yield of a crop of a county i in a year t, measured in bushels per acre, and the functional regressor $(X_{it}(s))_{s\in[0,40]}$ is a temperature curve, representing the crop exposure to temperatures between 0°C and 40°C during the growing season. Following Schlenker and Roberts (2009), the temperature curves are computed at discrete points $s_j \in \{0, 1, \ldots, 40\}$ as $X_{it}(s_j) = \Phi_{it}(s_j + 1) - \Phi_{it}(s_j)$, where $\Phi_{it}(s_j)$ is the length of time (measured in days) the crop was

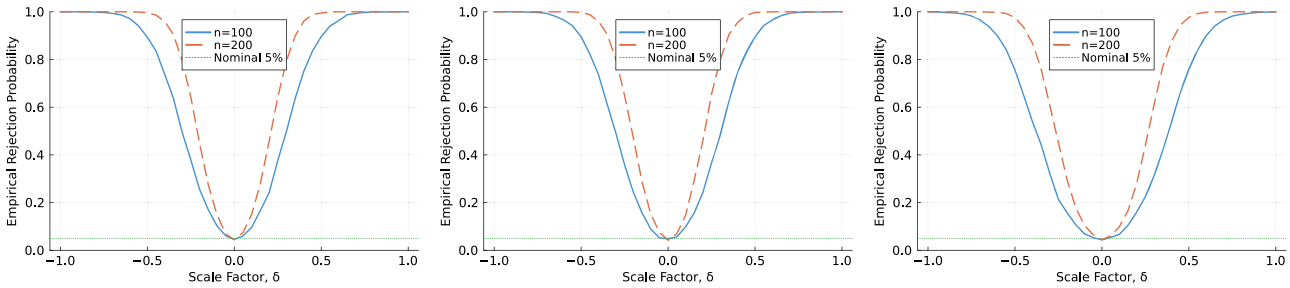
[7] We find that for these models the fitting error does not substantially decrease after $m = 10$ components.



(a) Model 1: Histograms (b) Model 2: Histograms (c) Model 3: Histograms

(d) Model 1: Q–Q plot (e) Model 2: Q–Q plot (f) Model 3: Q–Q plot

Figure 2. Accuracy of asymptotic approximations. Top row: finite-sample distribution of W_n under H_0 (blue histogram) overlaid with its asymptotic limit (orange histogram). Bottom row: Q–Q plots comparing empirical quantiles of W_n with the corresponding asymptotic quantiles.



(a) Model 1 (b) Model 2 (c) Model 3

Figure 3. Simulated power curves. Power curves for the three models corresponding to sample size $n = 100$ (blue solid curve) and $n = 200$ (red dashed curve). The green dotted line indicates the 5% nominal significance level.

continuously exposed to temperature larger than s_j for county i in year t.

We focus on corn and soybeans, which are the two major crops grown in the United States. The dataset consists of fine-scale county-level crop yields and weather outcomes, spanning U.S. counties east of the 100th meridian from 1950 to 2020. We use the same set of controls as in Schlenker and Roberts (2009), namely: a constant, precipitation measured in mm from March through August, precipitation^2, county fixed effects, and a state-specific quadratic time trend to capture technological change. The crop yields Y and the temperature curve X are regressed on these controls to obtain the residuals, which are subsequently used for the functional data analysis.

The slope coefficient is then estimated using: (a) our functional PLS estimator; and (b) a highly parameterized least-squares estimator with a step function approximation as in Schlenker and Roberts (2009). The latter fits a separate temperature effect for each 3°C bin from 0°C to 40°C, hence, it involves 13 parameters. On the other hand, our early-stopping rule finds $\hat{m} = 4$ functional PLS components both for corn and soybeans; see Appendix Section S.5 for more details on the implementation.

Figure 4 displays the estimated functional slope coefficient β corresponding to our functional PLS (red solid curve) and step function approximation (black dashed curve) for corn and soybeans. We find that the critical temperature after which the crop yields start declining is around 29°C–30°C which is similar to findings reported in Schlenker and Roberts (2009).

Lastly, we look at how the nonlinear temperature effects have changed over time. Figure 5 reports the estimated functional slope coefficient splitting the data into three subsamples: 1950–1973 (blue dotted curve), 1974–1997 (red dashed curve), and 1998–2020 (green solid curve). The results indicate that the negative temperature effects were larger during 1950–1973 compared to the most recent 22 years, especially for the extreme temperatures. The mitigation of extreme temperature effects may come from two sources: adaptation and CO_2 fertilization. CO_2 fertilization effects observed in our sample are likely to be small; see Nordhaus (2013) who argues that doubling the atmospheric concentration of CO_2 would increase crop yields by 10%–15%

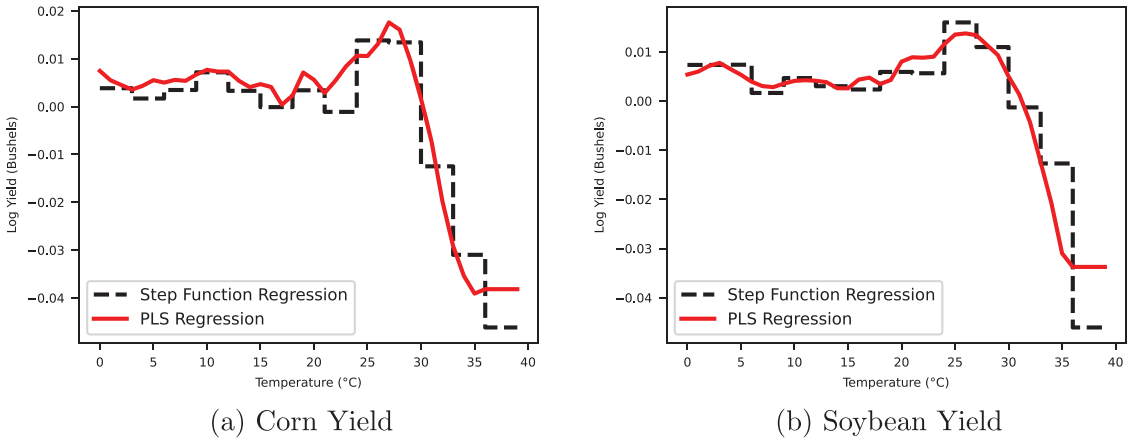


Figure 4. Nonlinear temperature effects. The figure shows the estimated nonlinear relationship between temperature and crop yields fitted using functional PLS (red solid curve) and step function approximation (black dashed curve).

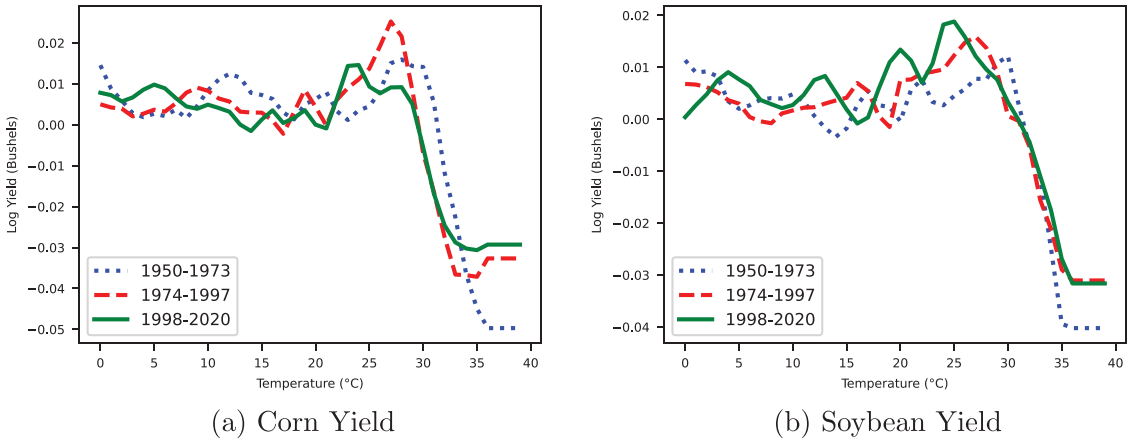


Figure 5. Adaptation. The figure plots the estimated nonlinear relationship between temperature and crop yields for three subsamples: 1950–1973, 1974–1997, and 1998–2020.

only. In contrast, the adaptation effect is likely to dominate over time. It can be attributed to the actions taken by farmers, such as adjusting the sowing and harvesting dates to maximize yields, using more resilient crops, or building efficient irrigation systems. Our results, therefore, suggest some evidence of adaptation in U.S. agriculture which has been reported in Burke and Emerick (2016) without properly accounting for nonlinearities. However, our confidence sets are too wide to draw conclusive inferences once statistical uncertainty is accounted for because there are only a few observations of extreme temperatures in the data.

7. Conclusions

This article reframes functional partial least squares (PLS) as a conjugate-gradient regularization scheme for ill-posed functional linear regression. Under mild conditions we prove that the resulting estimator achieves the minimax convergence rate (up to a logarithmic factor) and introduce a fully data-driven early-stopping rule that adapts automatically to the unknown degree of ill-posedness. We also propose a novel test and a confidence set for the functional coefficient β.

Monte Carlo evidence indicates that the estimator matches—or exceeds—the estimation and prediction accuracy of leading alternatives, particularly functional PCA, while requiring substantially fewer components. We also find that the early stopping rule consistently adjusts to the underlying signal-to-noise structure.

Applied to seven decades of U.S. county-level crop data, we find that the adverse impact of high temperatures on corn and soybean yields has weakened since the 1950s, hinting at adaptive responses in U.S. agriculture. Because observations of extreme heat are sparse, however, firmer conclusions will require richer data or refined inference procedures.

Supplementary Materials

Supplementary material contains all proofs, comparison to alternative estimators, and additional results. Moreover, we provide a replication package, available at *https://github.com/ababii/functional_pls_jasa_replication*, which reproduces all results in the paper.

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The authors report there are no competing interests to declare.

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